GENAI BASICS • LESSON 06 • MODULE 2: MULTIMODALITY

Speech and Audio

Voice-over, sound and music for a video, and how to check the result

Text-to-speech, speech-to-text, voice agents, music. ElevenLabs, OpenAI, Google, Suno, open-weight models. Data as of October 2026.



EXAMPLE

The L05 courier video with voice-over, sound and music



3 lines

9.2 s of speech from an **open-weight** text-to-speech model (downloadable, runs locally) on an ordinary laptop

3 tracks

voice + the sound Veo generated with the video + music

-14.0 LUFS

loudness in Loudness Units relative to Full Scale (LUFS), as the ear hears it: 0 = maximum, -14 = streaming norm, -23 = TV

LESSON PIPELINE 1 Script 2 Spoken text 3 Voice 4 Effects and music 5 Mix 6 Check

WHERE TO GENERATE

Three access layers for audio models: chat app, web studio, application programming interface (API)

Chat apps

voice and music inside a chat

ChatGPT · Gemini app

CONTROL	voice mode and a prompt; little control over voice and length
BILLING	subscription
USE FOR	trying things, music drafts
START WITH	**Gemini app (Lyria 3.5)**

Web studios

dedicated audio interface

ElevenLabs · Suno · Google AI Studio

CONTROL	voice library, settings, sound effects, music, dubbing
BILLING	credits; small free allowance
USE FOR	voicing a video, homework
START WITH	**ElevenLabs**

API

code and automation

OpenAI Audio · Gemini API · Google Cloud / Azure Speech · ElevenLabs API · Deepgram / AssemblyAI

CONTROL	everything as request parameters; batch jobs, live audio
BILLING	per character, token or minute
USE FOR	pipelines, products, agents
START WITH	**Gemini API**

A fourth path: open-weight models (model files you download and run on your own machine). Whisper (speech recognition), Kokoro and Piper (speech synthesis), MusicGen (music). $0 per minute, but hardware, setup and licences are yours. Our example was made this way.

TASK MAP

Four audio AI tasks: what goes in and what comes out

text → **voice**

Text-to-speech (TTS)

video voice-over, notifications, audiobooks. **Today: the courier narration.**

voice → **text**

Speech-to-text (STT)

subtitles, meeting notes, call search. Also called automatic speech recognition (ASR). **Today: checking the voice-over.**

voice → **voice**

Speech-to-speech

voice agents, live translation, dubbing. Replies in real time.

text → **music, effects**

Music and sound effects (SFX)

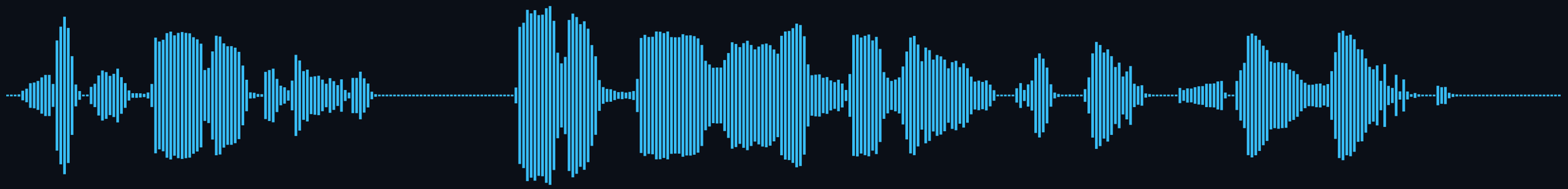
beds, jingles, effects. Plus the sound a **video model** makes by itself (L05).

HOW IT WORKS

How a text-to-speech model turns text into sound



LINE 1 OF OUR SCRIPT, REAL WAVEFORM · 3.9 S



How much data: one second of studio-quality audio is 48,000 numbers (Gemini returns 24,000). Models work with a compressed form: Gemini bills **25 audio tokens per second**, MusicGen takes **50 steps**.

What is not inside: a rulebook for reading. The model learned how text usually sounds, so it most often errs on **numbers, names and abbreviations**.

BRIDGE FROM L05

Audio inside a video model: what Veo 3.1 generated by itself

Shot 1, cafe prompt has no dialogue



▶ audio in the HTML version

HEARD IN VEO'S AUDIO

"Here's your order. Thank you. Have a good day."

The L05 failure clip prompt has no dialogue either



▶ audio in the HTML version

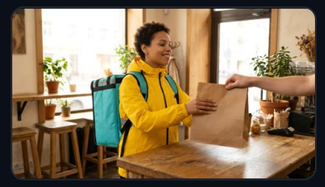
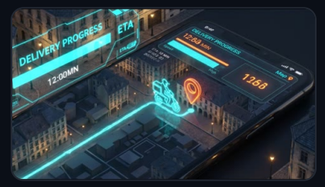
HEARD IN VEO'S AUDIO

"Continue straight for 200 meters, then your destination is on the left."

Provides: street noise, rain, footsteps, in sync with the picture. But you pay for it: Veo 3.1 on Vertex AI is $0.40/s with audio, $0.20/s without.

Does not provide: exact words, one voice across shots, separate tracks. The model **makes up lines** by itself.

Voice-over script: text, timing and a word budget for each shot

SHOT	TIME	VOICE-OVER	WORDS / LENGTH	SOUND EFFECTS
	0–8 s	“6:42. Order 417 is picked up, still hot.”	11 / 3.9 s	cafe: room noise, cups
	7.5–15.5 s	“Rain on the avenue. The route skips the bridge.”	9 / 2.7 s	rain, tyres on puddles
	15–23 s	“12 minutes to the door. Anna is on her way.”	10 / 2.7 s	soft UI sound

Music: one bed for all shots, lo-fi, no vocals

≈ 3.2 words/s

measured on our voice-over; plan for **2–2.5 words per second** to leave a pause

An 8 s shot ≈ 6 s of speech (about 1 s of silence at each cut) **× 2–2.5 words/s = 12–15 words.** If it does not fit, cut the text; do not speed up the voice.

Spoken text: numbers as words, style as a separate instruction

WHAT THE VIEWER SEES → WHAT THE MODEL READS

SCRIPT	SPOKEN TEXT	WHY
6:42	Six forty-two	a time, a score or a ratio?
417	four one seven	an order ID: digit by digit, not “four hundred seventeen”
12 min	twelve minutes	abbreviations may be read letter by letter

STYLE GOES IN A SEPARATE INSTRUCTION

```
Instruction: Calm, friendly dispatcher. Medium pace.
Short pause after each sentence.
Text: Six forty-two. Order four one seven is picked
up, still hot.
```

- **OpenAI:** a separate `instructions` field; **Gemini 3.8:** a separate `style` field, short sounds as tags: `<sigh>`.
- **ElevenLabs v3:** tags inside the text, e.g. `[whispers]`.
- **Pronunciation dictionary** for names and terms (ElevenLabs, Google, Azure).

Three sources of a voice: library, description, clone

Library voice

Input: pick from the service's library

✓ **covered by the service licence**

Gemini: 30 voices; OpenAI: 13; our example: Kokoro `af_heart`, Piper `dmitri`

Designed voice

Input: a text description: "young voice, slightly husky, fast pace"

! check it does not resemble a real person

ElevenLabs Voice Design, Gemini API voice design

Cloned voice

Input: recordings of a real person, from 3–10 s to 30+ min

✗ **only with the owner's written consent**

services require a recorded consent phrase or a voice check

Before you commit: run the hardest words of the script (names, numbers) through the voice. Reject a voice that mispronounces them.

Sound effects and music: prompt, length and the track licence

Our music bed MusicGen-small · 23 s



▶ audio in the HTML version

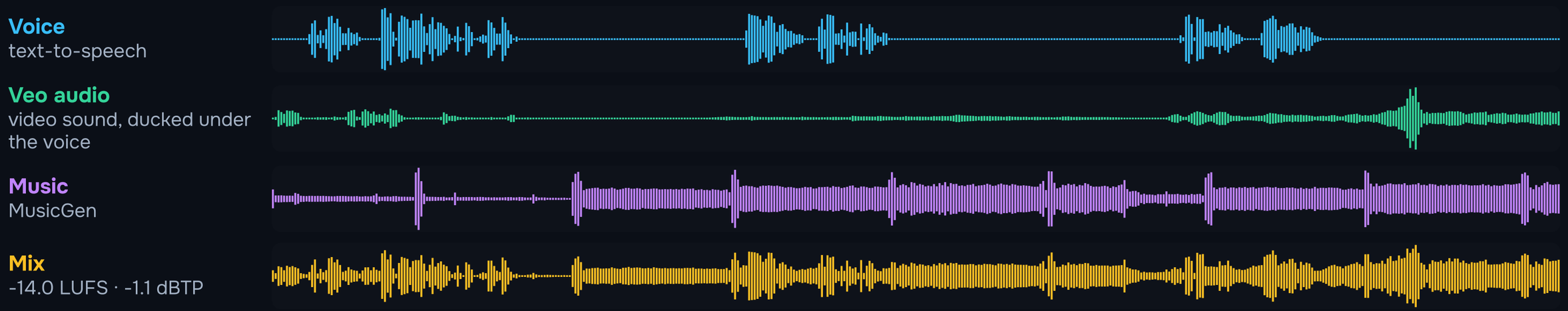
PROMPT

```
calm lo-fi hip hop beat, soft electric piano, light rain mood, 85 bpm, no vocals
```

> MusicGen weights: **Creative Commons Attribution-NonCommercial (CC BY-NC 4.0)** licence. Fine for a lesson, not for a client.

SERVICE	LENGTH	COMMERCIAL USE
Suno v6 songs	song	paid plans only
Lyria 3.5 Gemini app, Gemini API	~2 min	not stated; SynthID watermark on every track
ElevenLabs Music music	3 s – 5 min	“nearly all”, per plan
ElevenLabs Sound Effects sound effects	≤ 30 s	per plan

Mixing: voice on top, background ducked, loudness normalised



- **Ducking:** the background drops automatically while the voice speaks.
- **Loudness:** -14 Loudness Units relative to Full Scale (LUFS) for Spotify-style streaming, -23 LUFS for broadcast (European Broadcasting Union standard EBU R128).
- **Peak:** true peak, the highest point of the wave including between samples; no higher than -1 decibel (dBTP), otherwise MP3 or AAC encoding distorts it.

No code, Audacity (free): 1) import the video sound, music and lines; 2) background: *Effect › Auto Duck*; 3) *Mix and Render* → *Loudness Normalization* -14 LUFS → *Limiter* -1 dB; 4) WAV into CapCut or Clipchamp, mute the clip's own sound.

With code: ffmpeg (sidechaincompress + loudnorm), full command in make_audio.py.

Checking the voice-over with speech-to-text

SCRIPT

6:42. Order 417 is picked up, still hot. Rain on the avenue. The route skips the bridge. 12 minutes to the door. Anna is on her way.

WHAT WHISPER HEARD (SMALL MODEL, LAPTOP)

~~Here's your order. 642.~~ Order **417** is picked up, still hot. Rain on the Avenue, the route skips the bridge. **12 minutes** to the door, **Anna** is on her way.

- **“Here’s your order.”**: the mix: Veo’s invented line leaks through
- **“642” instead of “6:42”**: speech-to-text formatting; the voice said it correctly

18.5%

word error rate (WER) = (substitutions + deletions + insertions) / words in the script

3 / 4 → 4 / 4

critical terms correct: automatic → after listening (time, order number, minutes, name)

No code: ElevenLabs Speech to Text, Google AI Studio (prompt “Transcribe verbatim. Do not correct mistakes.”), Buzz (Whisper desktop app).

ARCHITECTURE

Three speech system architectures: file, stream, voice agent

Batch file
subtitles, meeting notes, our check

recording → transcription → text with timecodes

`seconds to minutes`
waiting is fine

Streaming
live captions, meeting notes

microphone → short chunks → partial → final text

`under a second`
partial text may change

Speech-to-speech (realtime)
voice agent, translation

voice → end of turn? → model + tools → voice

`seconds matter`
interruptions, silence, latency

A voice agent, two ways: a chain “speech-to-text → language model → text-to-speech” (easier to debug) or one speech-to-speech model (faster, more natural). We build agents in module 4.

MARKET • OCTOBER 2026

Audio models by task: commercial and open-weight

Text-to-speech (TTS) text → voice	ElevenLabs v4 / v3 · Gemini 3.8 Flash TTS · OpenAI gpt-4o-mini-tts (until 6 Jan 2027) · **Kokoro-82M** Apache-2.0 · **Qwen3-TTS** Apache-2.0
Speech-to-text (STT, ASR) voice → text	OpenAI gpt-transcribe · ElevenLabs Scribe v2 · AssemblyAI Universal-3.5 Pro · **Whisper large-v3-turbo** MIT · **Qwen3-ASR** Apache-2.0
Speech-to-speech voice agents, translation	OpenAI gpt-realtime-2.1 · Gemini Live API · ElevenLabs Agents
Music and sound effects text → music, effects	Suno v6 · Lyria 3.5 · ElevenLabs Music / Sound Effects · **MusicGen** CC BY-NC

Speech-to-text accuracy (English): best commercial ≈ 4–5% word error rate, best open ≈ 5–6%, Whisper large-v3 ≈ 7.8%. Hugging Face Open ASR Leaderboard, 9 October 2026.

Model lifetimes are short: OpenAI shuts down tts-1 and gpt-4o-mini-tts on 6 January 2027, and whisper-1 and gpt-4o-transcribe on 26 February 2027. Keep the model name in configuration, not in code.

Green border: open-weight model and its licence. **MIT, Apache-2.0:** commercial use allowed, keep the licence notice. **CC BY:** commercial use with credit. **CC BY-NC:** non-commercial only. Voices and datasets can have their own licences.

BREAK

5 minutes for coffee and resting your ears

05:00

Next: code, cost, acceptance, risks and homework.

REFERENCE • GOOGLE-GENAI + WHISPER

Code: text-to-speech with Gemini and a speech-to-text check with Whisper

```python
from google import genai
from google.genai import types

client = genai.Client()  # key from GEMINI_API_KEY
r = client.models.generate_content(
    model="gemini-3.8-flash-tts",
    contents=types.Content(role="user", parts=[types.Part(
        text="Six forty-two. Order four one seven.",
        speech_metadata=types.SpeechMetadata(
            style="calm, friendly dispatcher"))]),
    config=types.GenerateContentConfig(
        response_modalities=["AUDIO"],
        speech_config=types.SpeechConfig(
            voice_config=types.VoiceConfig(
                prebuilt_voice_config=types.PrebuiltVoiceConfig(
                    voice_name="Kore")))))
audio = r.candidates[0].content.parts[0].inline_data  # already WAV
open("line1.wav", "wb").write(audio.data)
```

> **Synthesis:** the same idea as in L03, but the response is audio: a WAV file, 16-bit, 24 kHz, mono. The voice style goes in a separate `style` field: Gemini 3.8 reads the text aloud word for word.

CHECK, LOCAL AND FREE

```python
from faster_whisper import WhisperModel
m = WhisperModel("small")
segs, _ = m.transcribe("mix.wav")
print(" ".join(s.text for s in segs))
```

Gemini application programming interface (API), October 2026. The full lesson pipeline: `deck_v2/make_audio.py`.

COST

Audio cost: per second, per minute, per hour

Conversion: **1 minute of speech ≈ 1,000 characters** (ElevenLabs' own rule). List prices as of 12 October 2026.

MODEL · BILLED AS	1 S	1 MIN	1 HOUR
Gemini 3.8 Flash TTS Text-to-speech · $9 / 1M audio tokens	$0.00022	$0.0135	**$0.81**
ElevenLabs v3 / v4 Text-to-speech · $0.08 / 1K characters	$0.00133	$0.08	**$4.80**
OpenAI gpt-transcribe Speech-to-text · $0.0045 / min	$0.00007	$0.0045	**$0.27**
ElevenLabs Scribe v2 Speech-to-text · $0.22 / hour	$0.00006	$0.00367	**$0.22**
OpenAI gpt-live-1 Speech-to-speech · $0.05 / min + model	$0.00083	$0.05	**$3.00**
ElevenLabs Music Music · $0.15 / min	$0.0025	$0.15	**$9.00**

≈ 1 cent

voicing our video: 154 characters on ElevenLabs v3; $0 for us on open models

$4.80 = 12 s

an hour of ElevenLabs voice costs as much as 12 seconds of Veo 3.1 video (L05: $0.40/s)

ACCEPTANCE

Acceptance checklist for a voice-over

The same items are the homework rubric. On the right, the real result for our version.

1	**Words** match the script: speech-to-text + critical terms list	✗
2	**Pronunciation** of names, numbers, terms: checked by ear	✓
3	**Timing:** each line fits its shot and is not cut by a transition	✓
4	**Mix:** voice is clear, background ducked, no stray speech	✗
5	**Loudness** −14 ± 1 LUFS; true peak ≤ −1 dB (dBTP)	✓
6	**Rights:** voice and music licence, consent, “AI-generated” label	!

-14.0 LUFS

peak -1.1 dBTP. No code: Youlean Loudness Meter or the DaVinci Resolve loudness meter; read Integrated and True Peak

> **Rejected:** Veo’s line “Here’s your order” leaks into the start. Fix: mute the shot 1 audio for 0–3 s, then repeat item 1.

Homework evidence: 1 → transcript.txt and critical-terms table · 2, 4 → by-ear notes with timecodes · 3 → script.md · 5 → meter screenshot · 6 → licence table and “AI-generated” label

RISKS

Risks and voice consent: cloning, labelling, law

Library or designed voice
allowed under the service licence, with an “AI-generated” label

Your own voice, a colleague’s clone
written consent: purpose, where, how long, how to withdraw

A client’s, manager’s or public figure’s voice
not without written consent and legal review; voices are used for fraud

HOW SERVICES CHECK CONSENT FOR A CLONE

- **OpenAI, Google:** the voice owner records a fixed consent phrase.
- **ElevenLabs Professional:** the owner reads a verification text aloud; Instant: a checkbox only.
- **Azure:** clones by application only; the consent recording is matched to the voice.
- **Open-weight models** (Qwen3-TTS, Chatterbox): clone from 3–10 s, **no checks at all**.

EU AI Act, Art. 50: from 2 August 2026, synthetic audio and deepfakes must be labelled; fines up to €15M or 3% of turnover. **OpenAI** requires telling listeners the voice is AI-generated. **Google** embeds an invisible SynthID watermark in audio from its own models.

PRACTICE

Homework 6: voice your L05 video



Your L05 video

Track A (product) or B (courier, Glovo case). If you have no video, use the lesson's courier video. No sound in your video: build the background from effects or music.

```
genai-homeworks/L06/
├── script.md       # text, spoken text, timing
├── audio/          # lines, bed, mix
├── final_voiced.mp4
├── transcript.txt # speech-to-text output
├── audio_spec.md
└── audio_qa.md
```

1. **Script**: 3 lines, word budget, critical terms → `script.md`
2. **Spoken text**: numbers and names as words, a style instruction
3. **Voice** in two no-code services (e.g. ElevenLabs and Google AI Studio), keep the better one
4. **Background**: video model audio, effects or music, with its licence
5. **Mix** into `final_voiced.mp4` : -14 ± 1 LUFS, a meter screenshot
6. **Check**: `transcript.txt` , slide 18 checklist → `audio_qa.md` , cost → `audio_spec.md`

Reviewer: **ihar_rubanovich@epam.com**. Your own voice or library voices only; do not clone anyone else's voice.

SUMMARY

Lesson summary

1. **Script with a word budget**
   2–2.5 words per second, a second of silence at each cut
2. **Spoken text kept separate**
   numbers, times, names as words; style as a separate instruction
3. **A voice with a licence and consent**
   a clone only with the owner’s written consent
4. **Background with a known licence**
   video audio, effects or music; MusicGen and free Suno are non-commercial
5. **Mix to −14 LUFS**
   voice on top, background ducked, true peak ≤ -1 dB (dBTP)
6. **Check with speech-to-text and by ear**
   word error rate + critical terms



Lesson 07: GitHub, Microsoft 365 and Copilot. From generating content to AI inside everyday work tools.